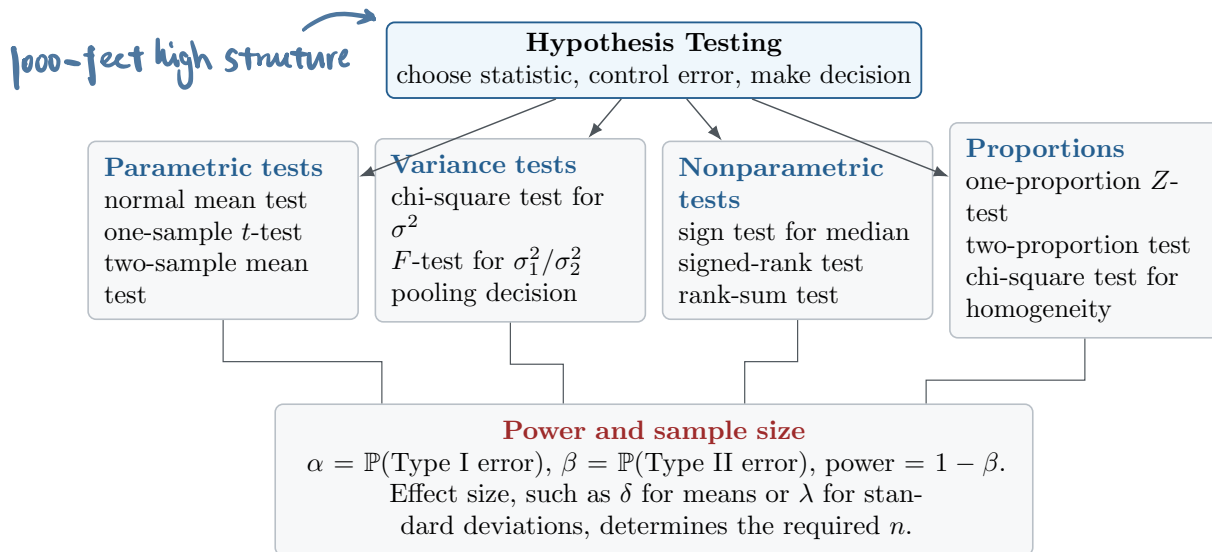


Lecture 16: Formula Summary Part II

1 Concept Map

The following concept map summarizes the main ideas of hypothesis testing, including parametric tests, variance tests, nonparametric tests, and proportion tests¹



¹The map should be read from top to bottom. Hypothesis testing is the common framework: write H_0, H_1 , choose a statistic, set a significance level, and decide whether the observed statistic falls in the rejection region. The middle row separates the main families of tests. Parametric tests use stronger distributional assumptions, often normality; variance tests check or compare spread and can decide whether a pooled two-sample t -test is appropriate. When normality is questionable, rank-based nonparametric tests replace raw values by signs or ranks. Proportion tests use binomial or large-sample normal approximations for categorical data. The bottom box connects to all tests: power and sample size are considered after choosing a test family, and α, β , effect size, and sample size describe how sensitive the test will be.

NHST

Fisher: Yes or NO question

2 Neyman-Pearson Testing and Power:

You must make a decision between the two

⇒ We DON'T CARE if H_1 is correct or not

Errors

Decision on H_0	H_0 is true	H_0 is false
Not reject	✓	Type II error
Reject	Type I error	✓

Powers

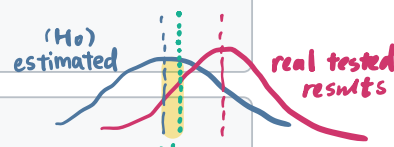
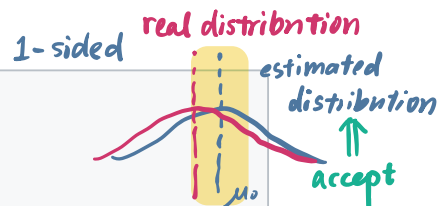
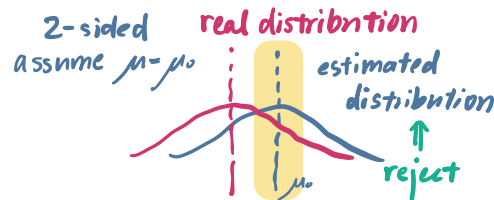
H_0 is correct but I reject

H_0 is wrong but I accept

$$\alpha = \mathbb{P}(\text{Type I error}),$$

$$\beta = \mathbb{P}(\text{Type II error}),$$

$$\text{power} = 1 - \beta.$$



How to run a hypothesis test

might be mean / median / variance / proportion

1. State H_0 and H_1 from the claim in the question.
2. Choose a statistic $T(X)$ whose distribution under H_0 is known.
3. Fix the level α and determine the rejection region R .
4. Compute T_{obs} and compare it with R .
5. $T_{\text{obs}} \in R \Rightarrow$ reject H_0 , $T_{\text{obs}} \notin R \Rightarrow$ do not reject H_0 .

Equivalently, using a P -value,

According to the following formulas provided in this doc.

$$P\text{-value} = \mathbb{P}_{H_0}\{\text{test statistic at least as extreme as } T_{\text{obs}}\},$$

$$P\text{-value} < \alpha \Rightarrow \text{reject } H_0.$$

$$\alpha = \mathbb{P}(X < z_\alpha)$$

$$\alpha = \mathbb{P}(-z_{\frac{\alpha}{2}} < X < z_{\frac{\alpha}{2}})$$

$$H_0: \theta = \theta_0$$

$$\Downarrow$$

use formula get $T_{\text{obs}} = \frac{\hat{\theta} - \theta_0}{\sigma}$

$\Downarrow$
compare with std distribution ($N(0,1), t_{n-1}, \chi_{n-1}^2, F_{n_1, n_2}$)

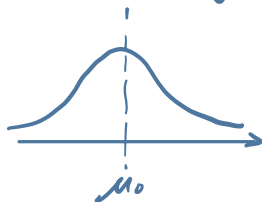
Assume mean over here (σ known)

①

$$H_0: \mu = \mu_0$$

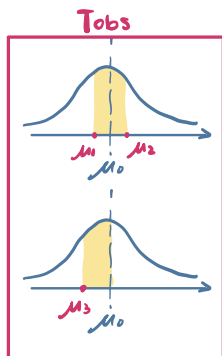
$$H_1: \mu > \mu_0$$

② (CLT) We know it won't be normally distributed d

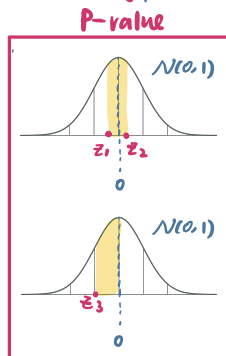


③ fix $\alpha = 0.05$

why? Because we can look up values directly from stats tables!



$$z = \frac{X - \mu}{\sigma}$$

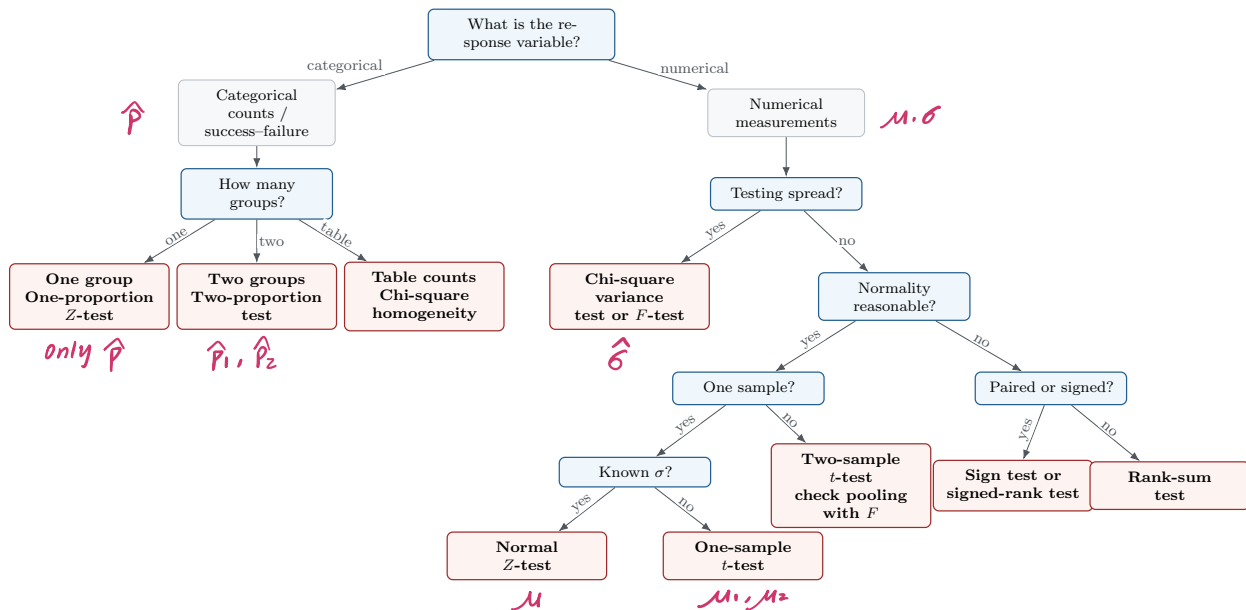


2

④ continue on calculation! get a value



3 Yes/No Decision Flowchart



Read the chart from the top. First identify the response variable, then follow the yes/no branches until a red test box is reached.

- If the response is categorical, use a proportion or count-based test: one group gives a one-proportion Z -test, two groups give a two-proportion test, and table counts or more than two groups give a chi-square homogeneity test.
- If the response is numerical and the target is spread, use a chi-square variance test for one variance or an F -test for comparing variances.
- If the response is numerical and the target is mean/median, first check whether normality is reasonable. With normality, one sample leads to either a normal Z -test when σ is known or a one-sample t -test when σ is unknown; two independent samples lead to a two-sample t -test, with pooling decided by the variance comparison.
- If normality is not reasonable, use a nonparametric test: paired or signed data lead to the sign test or signed-rank test, while two independent samples lead to the rank-sum test.

After choosing the test family, still check the exact alternative ($<$, $>$, $\neq$), the sample design, and the assumptions in the problem statement before setting the rejection region or computing the P -value.

Assume $x_1, \dots, x_n \sim \text{iid } N(\mu, \sigma^2)$

Normalize $z_i = \frac{x_i - \mu}{\sigma} \sim N(0, 1)$

$$\sum_{i=1}^n z_i^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma^2} \sim \chi_n^2$$

$\Rightarrow \chi^2$ -dist is the sum of several iid normal

4 Test Families from the Flowchart

4.1 Numerical Measurements

4.1.1 Testing Spread: Chi-Square Test for Variance

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$$

$\sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma^2} = \underbrace{\sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\sigma^2}}_{\frac{(n-1)S^2}{\sigma^2}} + \underbrace{\frac{n(\bar{x} - \mu)^2}{\sigma^2}}_{\text{determined by sample data}}$

$\frac{n(\bar{x} - \mu)^2}{\sigma^2}$ is determined by assumption known constant

Examples:

- Is the machine's standard deviation less than 1.5 mm?
- Is the filling process more variable than the specification allows?
- Has the process variance changed from the historical value σ_0^2 ?

Statistic

For a normal population,

$$S^2 := \frac{1}{n-1} \sum (\bar{x} - x_i)^2 \Leftarrow \text{observed sample data}$$

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2, \quad X^2 = \frac{(n-1)S^2}{\sigma_0^2}.$$

Rejection regions

$$\begin{aligned} H_1 : \sigma \neq \sigma_0 & \quad X^2 > \chi_{\alpha/2, n-1}^2 \text{ or } X^2 < \chi_{1-\alpha/2, n-1}^2, \\ H_1 : \sigma > \sigma_0 & \quad X^2 > \chi_{\alpha, n-1}^2, \\ H_1 : \sigma < \sigma_0 & \quad X^2 < \chi_{1-\alpha, n-1}^2. \end{aligned}$$

4.1.2 Power for Chi-Square Variance Tests

Examples:

- If the true standard deviation is 1.35 mm, what is the power of the left-tailed variance test?
- How many observations are needed to detect a smaller standard deviation with 90% power?
- If the real variation is only slightly smaller than σ_0 , is the test likely to reject?

Left-tailed alternative: $\lambda < 1$

For $H_1 : \sigma = \lambda \sigma_0 < \sigma_0$

$$\lambda^2 \chi_{\beta}^2 = \chi_{1-\alpha}^2 \iff \lambda = \sqrt{\frac{F_{\chi^2}^{-1}(\alpha)}{F_{\chi^2}^{-1}(1-\beta)}}.$$

Assume $H_0 : \sigma = \sigma_0, H_1 : \sigma < \sigma_0$

Let $\sigma = \lambda \sigma_0$

$$\gamma = \frac{(n-1)S^2}{\sigma^2} = \frac{(n-1)S^2}{\lambda^2 \sigma_0^2} \sim \chi_{n-1}^2 \Rightarrow \lambda^2 \gamma \sim \chi_{n-1}^2$$

$$\alpha = P_{H_0}(X^2 < c_\alpha), \text{ power} = P_\sigma(X^2 < c_\alpha) = P(\gamma < \frac{c_\alpha}{\lambda^2})$$

$$\begin{aligned} \alpha &= P(\text{Obs} \in R | H_0) \\ \beta &= P(\text{Obs} \notin R | \theta) \\ \text{power} &= 1 - \beta \end{aligned}$$

$$\text{power} = 1 - \beta = F_{\chi^2} \left(\frac{F_{\chi^2}^{-1}(\alpha)}{\lambda^2} \right).$$

Right-tailed alternative: $\lambda > 1$

For $H_1 : \sigma = \lambda \sigma_0 > \sigma_0$,

$$\lambda^2 \chi_{1-\beta}^2 = \chi_{\alpha}^2 \iff F_{\chi^2}^{-1}(1 - \alpha) = \lambda^2 F_{\chi^2}^{-1}(\beta).$$

$$\beta = F_{\chi^2} \left(\frac{F_{\chi^2}^{-1}(1 - \alpha)}{\lambda^2} \right), \quad \text{power} = 1 - \beta.$$

4.1.3 Normality Reasonable?

One Sample: Known σ , Normal Z-Test. Examples:

- Is the process mean different from 100 when the population standard deviation is known?
- Is the average fill weight less than the labeled amount when σ is known?
- Did a calibrated instrument produce a mean reading above the target value?

$$\begin{aligned} \bar{X} &= \sum_{i=1}^n X_i \\ E[\bar{X}] &= E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \mu \\ \text{Var}[\bar{X}] &= \text{Var}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} n \cdot \sigma^2 = \frac{\sigma^2}{n} \end{aligned} \quad \left. \vphantom{\sum_{i=1}^n} \right\} \bar{X} \sim \left(\mu, \frac{\sigma^2}{n}\right)$$

Statistic and distribution $\Rightarrow$ approximation for mean

For known σ^2 ,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \quad Z \sim \text{Normal}(0, 1) \quad (H_0 : \mu = \mu_0).$$

If H_1 is true with $\mu = \mu_0 + \delta$, then

$$Z \sim \text{Normal}\left(\frac{\delta\sqrt{n}}{\sigma}, 1\right).$$

Approximate sample size

For testing $H_0 : \mu = \mu_0$ against an effect size $|\mu - \mu_0| = \delta$ ³

$$\text{two-sided:} \quad n \approx \left(\frac{(z_{\alpha/2} + z_{\beta})\sigma}{\delta} \right)^2,$$

$$\text{one-sided:} \quad n \approx \left(\frac{(z_{\alpha} + z_{\beta})\sigma}{\delta} \right)^2.$$

Assume $H_0 : \mu = \mu_0$, $H_1 : \mu > \mu_0$
 $\alpha = P(Z > z_{\alpha})$

$$\beta = P_{\mu}(Z \leq z_{\alpha}) = \Phi(z_{\alpha} - \frac{\delta\sqrt{n}}{\sigma})$$

$$\Rightarrow -z_{\beta} = z_{\alpha} - \frac{\delta\sqrt{n}}{\sigma}$$

as we only denote z_{α}, z_{β} to be positive.

²The notation χ_p^2 means the upper-tail chi-square critical value used in lecture notes; the equivalent CDF notation is written beside it to avoid ambiguity.

³Here δ is the smallest practically meaningful difference. Larger δ or smaller σ requires fewer observations.

$$V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2 \Rightarrow \frac{S}{\sigma} = \sqrt{\frac{V}{n-1}}$$

When we do not know variance exactly, we use $S = \sqrt{\frac{1}{n-1} \sum (\bar{x} - x_i)^2}$.

One Sample: Unknown σ , t-Test. Examples:

- Is the average battery life greater than 10 hours when σ is unknown?
- Is the mean exam score different from last year's average?
- Is the mean tensile strength below the required engineering standard?

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{\sigma}{S} \boxed{\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}} = Z$$

normal distributed

Statistic

For a normal population with unknown variance,

$$T_{n-1} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad (H_0: \mu = \mu_0).$$

$$= \frac{Z}{\sqrt{V/(n-1)}}$$

we know $\frac{Z}{\sqrt{V/a}} \sim t_a$
where $Z \sim N(0,1)$. $V \sim \chi_a^2$

Rejection regions

$$H_1: \mu \neq \mu_0 \quad \text{reject if } |T_{n-1}| > t_{\alpha/2, n-1},$$

$$H_1: \mu > \mu_0 \quad \text{reject if } T_{n-1} > t_{\alpha, n-1},$$

$$H_1: \mu < \mu_0 \quad \text{reject if } T_{n-1} < -t_{\alpha, n-1}.$$

Power ingredients

$$t = \frac{Z + \delta\sqrt{n}/\sigma}{W}, \quad Z \sim \text{Normal}(0, 1), \quad (n-1)W^2 \sim \chi_{n-1}^2.$$

$$\text{noncentrality parameter} = \frac{\delta\sqrt{n}}{\sigma}, \quad d = \frac{|\mu - \mu_0|}{\sigma}.$$

Two Samples: t-Test. Examples:

- Do two teaching methods produce different average exam scores?
- Does a new drug change the mean recovery time compared with the old drug?
- Are the mean lifetimes of two brands of components different?

Estimator and sampling distribution

$$\widehat{\mu_1 - \mu_2} = \bar{X}_1 - \bar{X}_2.$$

For independent normal samples,

$$\bar{X}_1 - \bar{X}_2 \sim \text{Normal}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

normal distributions X, Y ,

$$E[X+Y] = E[X] + E[Y]$$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

$$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(-Y)$$

$$= \text{Var}(X) + (-1)^2 \text{Var}(Y)$$

pooled test: if $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$(n_1-1)S_1^2 = \sum_{i=1}^{n_1} (X_i - \bar{X})^2$$

$$(n_2-1)S_2^2 = \sum_{i=1}^{n_2} (Y_i - \bar{Y})^2$$

$$\Rightarrow S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} \Rightarrow T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

$\Rightarrow$ we have a $(n_1-1) + (n_2-1) = n_1 + n_2 - 2$
degree of freedom

4.1.4 Normality Not Reasonable

Paired or Signed Data: Sign Test for the Median. *Examples:*

- *Is the median die roll greater than 3.5?*
- *In paired data, are more after-values than before-values positive?*
- *Is the median customer rating above a neutral score without assuming normality?*

Signs

For a continuous distribution with median M ,

$$\star \quad \mathbb{P}(X < M) = \mathbb{P}(X > M) = \frac{1}{2}.$$

$$Q_+ = \#\{X_k : X_k - M_0 > 0\}, \quad Q_- = \#\{X_k : X_k - M_0 < 0\}.$$

At the boundary $M = M_0$ ⁴

$$Q_+ \sim \text{Binomial}\left(n, \frac{1}{2}\right), \quad Q_- \sim \text{Binomial}\left(n, \frac{1}{2}\right).$$

Rejection rules

$$H_1 : M > M_0 \quad \text{reject for small } Q_- : \mathbb{P}(Q_- \leq q_- \mid M = M_0) < \alpha,$$

$$H_1 : M < M_0 \quad \text{reject for small } Q_+ : \mathbb{P}(Q_+ \leq q_+ \mid M = M_0) < \alpha,$$

$$H_1 : M \neq M_0 \quad \text{reject if } \mathbb{P}(\min(Q_-, Q_+) \leq k \mid M = M_0) < \alpha/2.$$

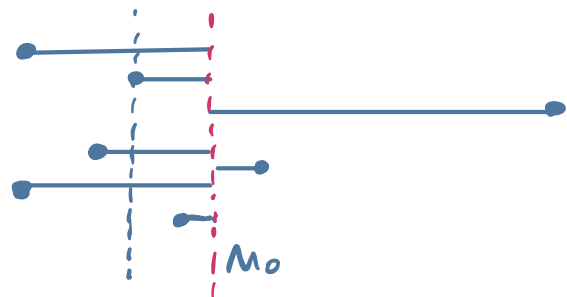
Paired or Signed Data: Wilcoxon Signed-Rank Test. *Examples:*

- *Is the median paired before-after change positive, using both signs and ranks?*
- *Did a training program improve scores when the paired differences are roughly symmetric?*
- *Are the signed deviations from a target centered at zero?*

Rank sums

$$W_+ = \sum_{R_i > 0} R_i, \quad |W_-| = \sum_{R_i < 0} |R_i|, \quad W = \min(W_+, |W_-|).$$

⁴Observations exactly equal to M_0 are usually discarded in the sign test; the effective sample size becomes the number of nonzero signs.



$$E[W^+] = E\left[\sum_{k=1}^n k B_k\right] = \sum_{k=1}^n k \underbrace{E[B_k]}_{\text{symmetry}} = \frac{1}{2} \sum_{k=1}^n k$$

$$\text{Var}(W^+) = \sum_{k=1}^n k^2 \text{Var}(B_k) = \frac{1}{4} \sum_{k=1}^n k^2$$

Distribution under symmetry

$$B_k = \begin{cases} 1, & \text{rank } k \text{ is positive,} \\ 0, & \text{rank } k \text{ is negative,} \end{cases}$$

$$B_k \stackrel{\text{iid}}{\sim} \text{Bernoulli}\left(\frac{1}{2}\right).$$

$$\text{Var}(B_k) = p(1-p) = \frac{1}{4}$$

$$E(B_k) = p = \frac{1}{2}$$

$$W_+ = \sum_{k=1}^n k B_k, \quad \mathbb{E}(W_+) = \frac{n(n+1)}{4}, \quad \text{Var}(W_+) = \frac{n(n+1)(2n+1)}{24}.$$

$$W_+ \approx \text{Normal}\left(\frac{n(n+1)}{4}, \frac{n(n+1)(2n+1)}{24}\right).$$

5

* Exact distribution

$$\mathbb{P}(W_+ = k) = \frac{1}{2^n} \#\{\text{subsets of } \{1, \dots, n\} \text{ summing to } k\}.$$

The numerator is the coefficient of x^k in

$$(1+x)(1+x^2)(1+x^3) \cdots (1+x^n).$$

Two Independent Samples: Wilcoxon Rank-Sum Test. Examples:

- Do two independent treatments tend to produce different response levels without assuming normality?
- Are waiting times generally shorter at one service desk than another?
- Do two independent populations have different locations when outliers make a t -test questionable?

Rank-sum statistic

Pool the two independent samples, rank all observations, and let

W = sum of ranks in sample 1.

Under H_0 , for sample sizes m and n ,

$$\text{Var}(W) = m \frac{(m+n-m)}{m+n-1} \cdot \frac{(m+n)^2 - 1}{12}$$

$$\mathbb{E}(W) = \frac{m(m+n+1)}{2}, \quad \text{Var}(W) = \frac{mn(m+n+1)}{12}.$$

For large samples,

$$Z = \frac{W - \mathbb{E}(W)}{\sqrt{\text{Var}(W)}} \approx \text{Normal}(0, 1).$$

⁵The signed-rank test uses both direction and magnitude rank. It is stronger than the sign test when the distribution is reasonably symmetric.

4.2 Categorical Counts / Success–Failure

4.2.1 One Group: Inference on One Proportion

Examples:

- Is the coin's chance of landing heads equal to 50%?
- Is the defect rate below 5%?
- Is the proportion of students who pass the exam greater than 80%?

Estimator and confidence interval

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}(np) = p$$

$$\hat{p} = \frac{X}{n}, \quad \hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}. \quad \text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{1}{n^2} \text{Var}(X) = \frac{p(1-p)}{n}$$

(CLT) when $n \rightarrow \infty$, $\hat{p} \sim N(p, \frac{p(1-p)}{n})$

Sample size

With prior estimate $\hat{p}$,

$$n \doteq \frac{z_{\alpha/2}^2 \hat{p}(1-\hat{p})}{d^2}. \quad z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq d$$

Without prior estimate⁶

$$\hat{p}(1-\hat{p}) \leq \frac{1}{4} \implies n \doteq \frac{z_{\alpha/2}^2}{4d^2}.$$

Test statistic

$H_0: p = p_0$ (You have already known the desired p_0 !)

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \quad np_0 > 5, \quad n(1-p_0) > 5.$$

we want it to fall in the range

4.2.2 Two Groups: Comparing Two Proportions

Examples:

- Are the pass rates different between two classes?
- Is the defect rate lower after changing the manufacturing process?
- Do two websites have different conversion rates?

⁶The maximum possible value of $p(1-p)$ is $1/4$, reached at $p = 1/2$. This gives the conservative sample size.

Estimator and confidence interval

(unpooled)

$$\widehat{p_1 - p_2} = \hat{p}_1 - \hat{p}_2 = \frac{X_1}{n_1} - \frac{X_2}{n_2}.$$

$$\hat{p}_1 - \hat{p}_2 \approx \text{Normal} \left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2} \right).$$

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}.$$

Tests

General null difference $(p_1 - p_2)_0$:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)_0}{\sqrt{\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2}}.$$

If $(p_1 - p_2)_0 = 0$, use the pooled estimate⁷

$$\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2} = \frac{X_1 + X_2}{n_1 + n_2}, \quad Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}.$$

* 4.2.3 Table Counts: Chi-Square Homogeneity

Examples:

- Do several regions have the same distribution of product preferences?
- Are political preferences distributed the same way across age groups?
- Do multiple treatments produce the same distribution of categorical outcomes?

Observed and expected counts

For an $r \times c$ table,

$$E_{ij} = \frac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{\text{grand total}}, \quad X^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

$$X^2 \sim \chi_{(r-1)(c-1)}^2 \quad (H_0 : \text{the group distributions are homogeneous}).$$

⁷Pooling is used only under the null hypothesis $p_1 = p_2$, because then both samples estimate the same common proportion.

